Huynh Ho

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Education

Gettysburg College Gettysburg, PA
B.S. Computer Science & Mathematics, Minor in Data Science, GPA: 3.71 *Expected May 2027*

Relevant Coursework: Operating Systems, Data Structures & Algorithms, Database Systems, AI, Probability & Statistics

Experience

Open Source Contributor, Backend Infrastructure May 2026 – Aug 2026
Kerno (GirlScript Summer of Code), Kubernetes observability tool in Go *1 merged PR*

- Replaced kubelet HTTP polling with a Kubernetes SharedIndexInformer scoped to the local node, fixing pod tracking that silently stopped on GKE/EKS clusters. Two in-memory indexes cut pod lookups from O(n) to O(1).
- Added exponential backoff (1s to 2 min) for API server failures and kept stale entries so pod enrichment keeps working in a degraded mode.

Open Source Contributor, Backend & Observability Jun 2026 – Present
Tether (FastCrest), edge-to-cloud AI deployment CLI, 83+ stars *3 merged PRs*

- Built request-ID middleware that stamps every API response with a UUID, so one request can be traced across a fleet. Merged after code review, with 5 unit tests.
- Fixed silent data loss on unclean shutdown by flushing pending writes at exit, with 6 tests.
- Restored a Prometheus metric that never emitted in production and fixed a double-counting bug.

Teaching Assistant, Calculus I Aug 2024 – Dec 2024
Gettysburg College *Gettysburg, PA*

- Led weekly problem-solving sessions for about 30 students and told the professor which topics they struggled with.

Projects

YouLecture: AI Study Environment | *Python, FastAPI, Claude API, Next.js, AWS* Live demo | 2026

- Built a 5-agent RAG pipeline that turns a YouTube URL into an outline, summaries, and flashcards.
- Deployed on AWS Lightsail with Docker and nginx. Parallelized slow stages and cached transcripts.
- Fixed YouTube HTTP 429 rate limiting on AWS with a rotating residential proxy.

Cancer to Hell: Clinical Research Tool | *Go, Anthropic Claude, PubMed, Next.js, SSE* Live demo | 2026

- Built three LLM agents (evidence, guideline, risk) running concurrently as Go goroutines and streaming over SSE. A guardrail checks each citation against live PubMed.
- Deployed on AWS Lightsail with Docker Compose.

Line-Following Robot: PicarX | *Python, Grayscale Sensors, State Machine* 2025

- Led a student team building a line-following robot. Tuned sensor thresholds for different lighting and added recovery logic to find the line again.

Technical Skills

Languages: Python, Go, Java, TypeScript, SQL
Tools: AWS, Docker, Kubernetes, Linux, Git, FastAPI, React, Prometheus
AI: Claude API, RAG, AI-assisted development with Claude Code